Boltzmann and, above all, Clausius. It was through the work of these investigators that the concept of a gas as something consisting of molecules in rapid motion—comparable in a way to a swarm of midges—gained a solid foundation in conformity with strict mathematical principles.

The year 1865 brought a new achievement of considerable importance: The first, and as yet only approximate, determination by Loschmidt of the size of atoms, and hence of the number of molecules contained in a given volume. Like Robert Mayer before him, Loschmidt investigated the internal friction of gases, and as a result of the preliminary researches of his precursor, he obtained a first clue to the size of an atom. His conclusions were still rather inaccurate, but he arrived at the right order of magnitude. The size of the atom has been known accurately for about forty years only. To give some idea of it, let it be mentioned that approximately ten million atoms placed next to each other in a straight line would form a line of one millimetre in length. Therefore, individual atoms are completely invisible, and it is impossible to observe them directly. They are infinitely smaller than those motes in a sunbeam which Democritus regarded as comparable with atoms with respect to order of magnitude.

The following years produced another step forward in the field of our knowledge of electricity. Through Faraday's discoveries, the existence of atoms of electricity had become a probability, but they were still known solely in association with atoms of the chemical elements, and not in a free state. Free atoms of electricity, atoms not bound to atoms of ordinary matter, were discovered by Hittorf in cathode rays, which are a consequence of electrical discharges in highly attenuated gases. Hittorf (1824–1914) studied the deflection of cathode rays in magnetic fields, and found that on the ground of the magnitude of this deflection it was possible to calculate the ratio of the charge to the mass of those particles moving in the cathode rays. As the mass of an individual atom had been known since Loschmidt's time, and as on the ground of Faraday's discoveries the size of the atoms of electricity could be computed to a fair approximation, the magnitude of the mass with which the atom of electricity was associated in its free state—in the cathode rays—was now determined in connection with the ratio just mentioned. As a result of more recent measurements, it is now known that this mass is about one 1,840th part of the mass of the lightest of all atoms, the hydrogen atom.

These atoms of free electricity are called *electrons,* a name first suggested by Stoney.

Of major importance is the fact that the wide range of masses like those of the atoms of the chemical elements is absent in the case of the atoms of electricity. Electrons are always associated with the same mass, which circumstance is in excellent conformity with the demand for homogeneity in nature. The view that electrons might be, in some way or another, constituent